

Lattice Proton Intercalation to Regulate WO₃-Based Solid-Contact Wearable pH Sensor for Sweat Analysis

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Sweat pH monitoring is a routine indicator in wearable biotechnology. The state-of-the-art wearable pH sensors mostly rely on organic materials but face the risk of biological toxicity. WO₃ is a typical H⁺-sensitive inorganic material with chemical stability, biocompatibility, and low cost but low sensitivity and slow response. Lattice H⁺ intercalation is herein proposed as an efficient approach that can greatly improve the sensitivity and selectivity of WO₃-based pH sensors. Specifically, lattice H⁺ intercalation can promote WO₃ from the monoclinic phase to cubic phase, which enhances the ion exchange capacity between WO₃ and H⁺. The resistance decreases more than two orders of magnitudes, which improves the interfacial charge transport. The occupancy of lattice H⁺ leads to ion exchange only with H⁺, thus increasing the H⁺ recognition. The intercalated H_xWO₃ exhibits much improved sensitivity, reversibility, and response time. Additionally, the H_xWO₃ is integrated with a solid reference electrode on a miniaturized chip for wearable sweat pH monitoring. The pH sensor exhibits good potential response even at curving over 270°. On-body sweat pH measurements show high accuracy compared with ex situ analyses. This work emphasizes the concept of lattice proton intercalation to regulate the H⁺ recognition of solid contacts.

1. Introduction

Measurement and control of hydrogen ion concentrations (pH) are essential for many chemical and biological applications.^[1] A wealth of health-related information can be revealed by testing pH values such as human sweat, saliva and other biological fluids.^[2–4] Likewise, pH value can also be as an important assessment in food processing^[5] and water analysis.^[6] At present, glass electrode is the most widely used pH sensor. However, the glass electrode has a few inherent drawbacks, such as high-temperature instability, acid and alkaline errors, high impedance and difficulties of miniaturization particularly,

which limits its further application in many circumstances.^[7,8] Typically, the intensively-focused electrochemical wearable sensors require flexible devices.^[9–14] The demanding of ion detection in small sample volumes and point-of-care testing motivates the development of miniaturized and wearable potentiometric sensors.^[15–19]

Solid-contact potentiometric pH sensors are a representative type of miniaturized devices that have been applied in wearable biosensing.^[20] Currently, the flexible pH sensors are focused on organic materials like polyaniline^[13] or proton ionophores.^[21] However, the biotoxicity for these sensing materials could be a risk for practical use. The polyaniline alone has less cytotoxic but the possible impurities like low-molecular-weight by-products could have significant carcinogenic effect of benzidine.^[22] The organic proton ionophores like tridodecylamine^[23] and 4-nonadecylpyridine^[24] are not only

costly, but also have the risk of leakage from polymer-based ion selective membranes upon skin exposure. Carbon nanotubes (CNTs) have also been used as a H⁺-sensitive material owing to their interactions between proton and surface functionalities of CNTs.^[25] However, the notable biotoxicity for the CNTs have been demonstrated in nanotoxicology.^[26]

Ion-sensitive metal oxides (MO_x) with micro- and nanostructures have been proven to be excellent candidates for the next generation of solid-contact pH sensors due to its high mechanical strength, high thermal stability and miniaturization.^[27] WO₃ is one of the most promising metal oxides due to its chemical stability,^[28,29] biocompatibility,^[30,31] and low cost.^[31,32] However, the sensitivity of the WO₃-based pH sensors is significantly influenced by its microstructure, porosity, surface homogeneity and crystalline structure. Typically, the WO₃ prepared by different or even similar methods discloses different pH sensitivities^[28,32–38] (Table S1, Supporting Information). Drenslar et al. reported that crystalline structure determined the pH-range and the sensitivity in compact WO₃ materials.^[35] Especially for the monoclinic phase, H⁺ can hardly be transported. Recently, Choi et al. demonstrated a sub-Nernstian response for WO₃ nanofiber and proposed an approach by using metal-oxide field-effect transistor as a differential amplifier for signal amplification.^[33] The output sensitivity increases ten-times higher than that without differential

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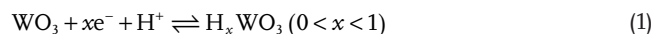
amplifier. Although great progress has been achieved, the specific conditions and complex operations turn into an obstacle for further application. Moreover, another significant challenge for the WO_3 -based pH sensors is the slow response. Therefore, there is a need to develop a simple and common strategy to further improve the sensing performance of WO_3 -based pH sensor.

Herein, we present an approach to improve the sensitivity and selectivity of WO_3 -based solid-contact pH sensor by one-step electrochemical proton lattice intercalation. After intercalation, the WO_3 reveals significant structure conversion from monoclinic phase to cubic phase, which improves proton exchange ability and decreases the resistance. The occupied proton sites also enhance the selectivity and other analytical performances, including reversible response, fast response time, and good anti-interferences. We used the proton intercalated H_xWO_3 to fabricate a wearable pH sensor. On-body sweat pH analysis also exhibits a high accuracy compared with ex situ analysis. These results demonstrate a simple strategy to regulate the pH sensing by ion intercalation.

2. Results

2.1. Lattice Proton Intercalation of WO_3

The pH sensing mechanism of WO_3 is dependent on the double injection/extraction of electrons and proton to the structure, which can be described by the reversible reaction



Significantly, WO_3 discloses ReO_3 -type structure at room temperature (Figure 1a). The difference is primitive cubic and conductive for ReO_3 , whereas WO_3 is monoclinic and semi-conductive.^[39] The low electrical conductivity leads WO_3 to a natural disadvantage in ion-to-electron transduction. Another difference lies between WO_3 and ReO_3 in their M-O-M bonds angles. ReO_3 has an angle of 180° , while WO_3 has angles between 165° and 179° (ref. [40]) (Figure 1a). This distortion is a consequence of the propensity for tungsten atoms to be displaced from the ideal cubic location in the WO_6 octahedral environments.^[39]

For this purpose, we applied a bias voltage (0 to -0.45 V) using linear sweep voltammetry (LSV) to the WO_3 in 0.05 M H_2SO_4 solution to proceed the reaction of Eq. (1) to form H_xWO_3 (Figure 1c). As shown in Figure 1d, WO_3 turns from yellow to dark purple after proton intercalation, which is a typical electrochromic process, indicating that H^+ was inserted into WO_3 lattice and W^{6+} was reduced to W^{5+} or lower valence state. The proton insertion of WO_3 causes the change in structure and properties. The monoclinic WO_3 becomes cubic hydrogen tungsten bronze (H_xWO_3) after protons intercalation. Moreover, protons are more easily to be inserted into a structure consistent with the existence of stoichiometric and crystalline phase (Figure 1b). This structural change is also accompanied with an increase in conductivity. Electrochemical impedance spectra (EIS) of WO_3 and H_xWO_3 are parallelly shown in Figure 1e,f. In the whole region of frequencies, the impedance of H_xWO_3 decreased more than two orders of magnitudes compared to WO_3 . Such a huge decrease in resistance suggests that the proton insertion greatly reduces the contact resistance and improves the charge transport

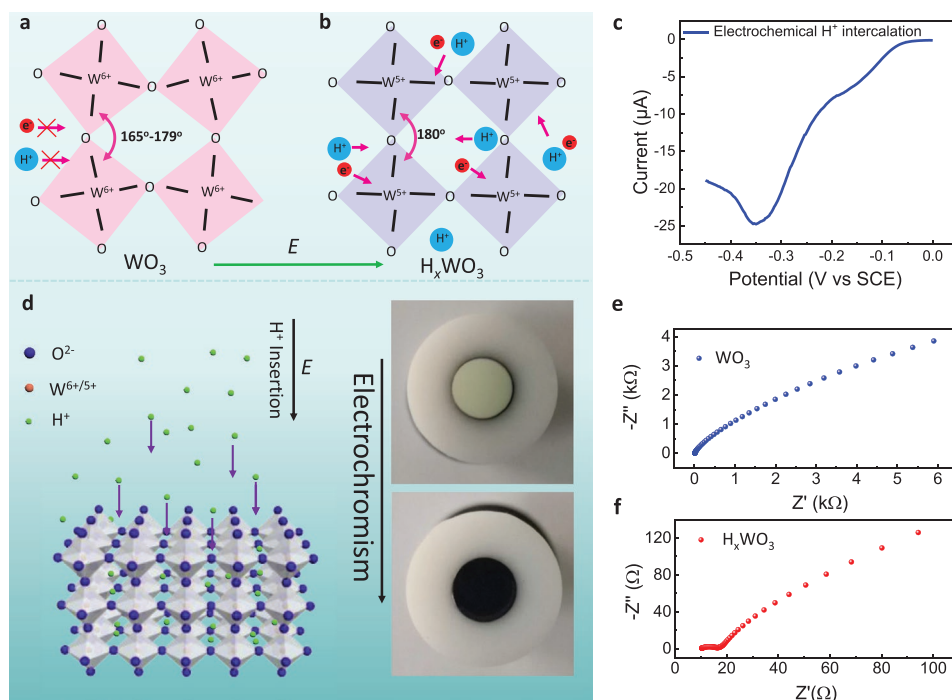


Figure 1. Lattice proton intercalation of WO_3 . a,b) Structures of WO_3 before and after proton intercalation (H_xWO_3). c) Electrochemical proton intercalation of WO_3 by LSV in 0.05 M H_2SO_4 with the scan rate of 0.5 mV s^{-1} . d) A schematic of proton inserting the lattice of WO_3 . The electrochromic of WO_3 film before and after H^+ intercalation. e,f) Electrochemical impedance spectra (EIS) of WO_3 and H_xWO_3 under open circuit potentials with an amplitude of 10 mV in 0.05 M H_2SO_4 . Frequency range: 1 M– 0.1 Hz.

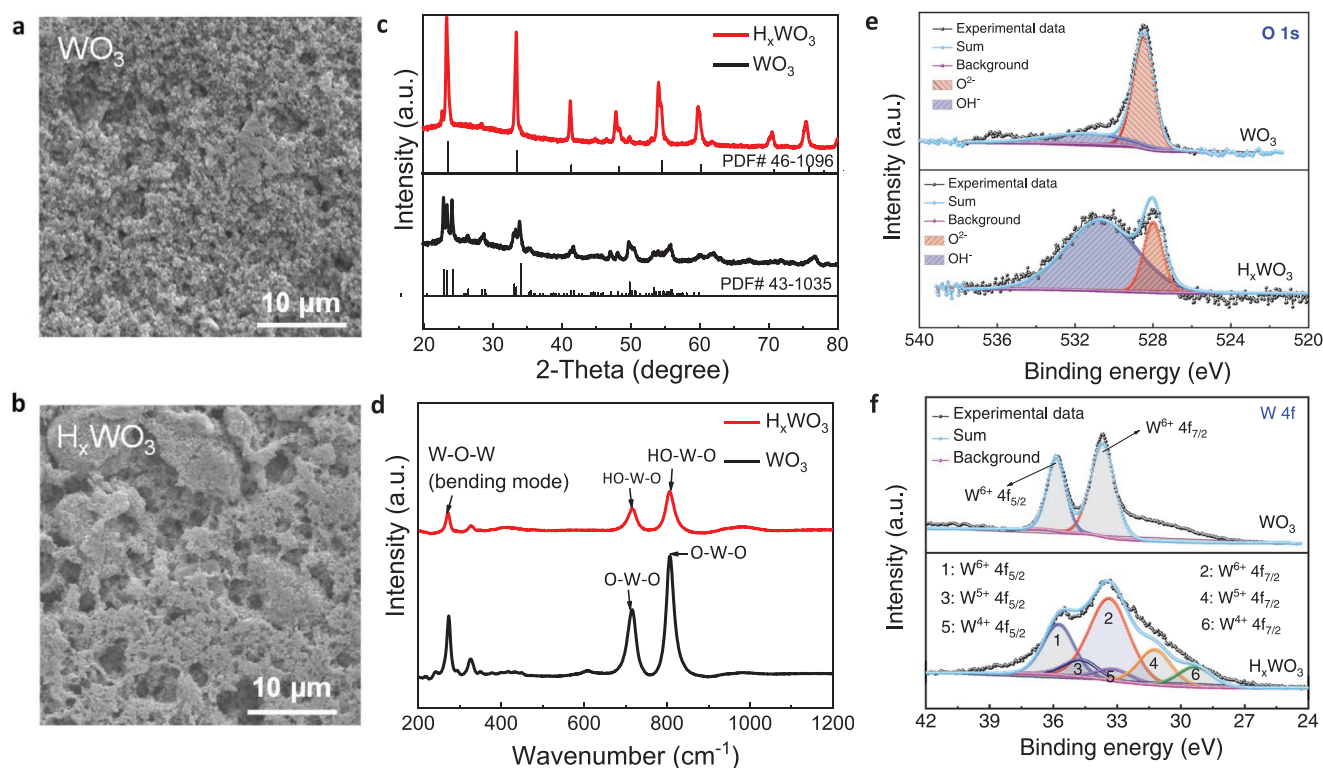


Figure 2. Structure comparison of WO_3 and H_xWO_3 . a,b) SEM images of WO_3 and H_xWO_3 . c) XRD patterns of WO_3 and H_xWO_3 . Corresponding PDF cards data are presented. d) Raman spectra of WO_3 and H_xWO_3 with the 532 nm laser. e,f) High-resolution XPS spectra of e) O 1s and f) W 4f for WO_3 and H_xWO_3 .

at solid contact interface. In the region of low frequencies, the proton diffusion Warburg resistance of WO_3 is much higher than H_xWO_3 , indicating that the intercalation of H^+ greatly facilitates the proton transport in the solid contact.

2.2. Structure Comparison of WO_3 and H_xWO_3

The morphology changes of WO_3 after proton intercalation were examined by scanning electron microscopy (SEM). WO_3 discloses a uniform distribution in granular form (Figure 2a; Figure S1a,c, Supporting Information) while H_xWO_3 shows porous channels with larger surface area (Figure 2b; Figure S1b,d, Supporting Information). This porous structure increases the effective sensing area of the membrane and the sites of charge distribution at the solid contact/solution interface, which is beneficial to improve the performances of the pH sensor.

After proton intercalation, the crystal structures were changed significantly and demonstrated by the X-ray diffraction (XRD) (Figure 2c). It is observed the WO_3 comprises crystalline monoclinic structure, while the diffraction peaks of H_xWO_3 are identical to the cubic phase. After intercalation, protons occupy the interstitial positions in the distorted lattice and are surrounded by nearly regular octahedrons of oxygen atoms. Compared to monoclinic WO_3 , the oxygen position changes, and displaced by the normal perovskite sites along $\langle 110 \rangle$, which prompts the crystal structure from monoclinic to symmetrical cubic phase.^[41] The Raman spectra of the WO_3 and H_xWO_3 are

parallelly shown in Figure 2d. As expected, Raman bands centered at 806 and 715 cm^{-1} correspond to the O-W-O stretching modes, which are the main characteristic peaks of WO_3 crystallites.^[42] Compared with WO_3 , the main bands for H_xWO_3 are broadened and weakened, due to the formation of W-OH bonds that weaken the initial O-W-O bonds and change the bond angles.^[43] The weakened bending mode of W-O-W band at 262 cm^{-1} further indicates the proton intercalation. Combined with the XRD results, the main bands of O-W-O bond in the crystalline WO_3 is highly sensitive to changes in crystal symmetry and lattice distortion.^[41,44] Overall the crystal structure of WO_3 changed from a monoclinic to a more symmetric cubic phase along with the insertion of hydrogen ion.

X-ray photoelectron spectroscopy (XPS) characterization reveals the surface chemical compositions of the WO_3 and H_xWO_3 . The core W 4f and O 1s confirm the oxidation valence states of W. Figure S2 in the Supporting Information shows four elements (F, W, O, and C) exist in WO_3 and H_xWO_3 . Among them, the F and C elements were introduced by the PVDF binder. The deconvolution peak of the O 1s spectrum can be resolved into two components of 529.8 and 532.3 eV attributed to the O^{2-} bond with wolfram and OH^- , respectively^[43] (Figure 2e). It is obvious that the binding energy of OH^- increases after H^+ intercalation. According to above results of XRD and Raman, both oxygen atom position and O-W-O bond changed, and H^+ attaches to oxygen atoms (OH^-),^[45] which leads to an increase of binding intensity. Figure 2f presents the XPS spectra of the W 4f doublets (33.7 and 35.8 eV) corresponding to W 4f_{7/2} and W 4f_{5/2} respectively, which is

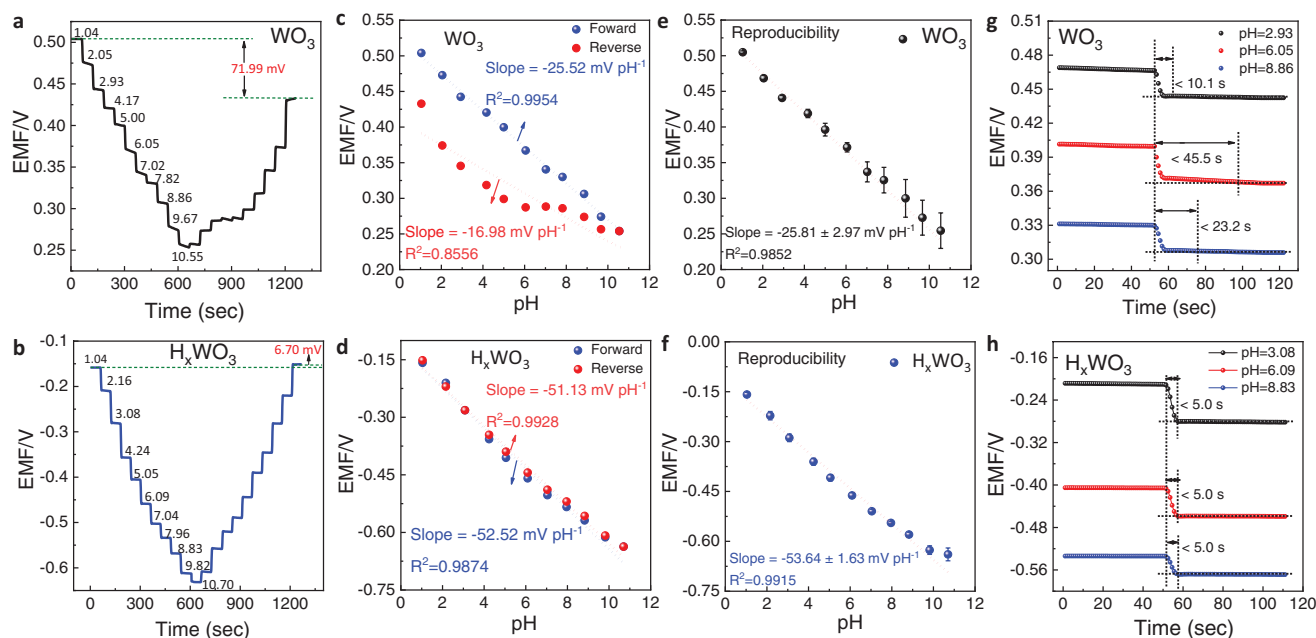


Figure 3. Potentiometric pH responses of WO_3 and H_xWO_3 . a,b) Examination of pH reversible responses for WO_3 and H_xWO_3 . c,d) Linear response curves of the cyclic tests. e,f) The reproducibility examination for three individual electrodes ($n = 3$). g,h) Response times for the WO_3 and H_xWO_3 .

consistent to oxidation valence of W^{6+} in WO_3 .^[46] For H_xWO_3 , W 4f spectrum was fitted with three spin-orbit doublets corresponding to the three different oxidation states of W atoms. The peak-differentiation-imitating analysis of H_xWO_3 indicates the presence amount of W^{6+} , W^{5+} , and W^{4+} was 65.33%, 20.4%, and 14.2%, respectively, which confirmed that the insertion of protons into the WO_3 lattice changed the oxidation state of W.^[47] By calculation, the average oxidation state of W was 5.51+, thus, the chemical formula of H_xWO_3 can be estimated as $\text{H}_{0.49}\text{WO}_3$.

2.3. Potentiometric Responses of WO_3 and H_xWO_3

After exploration of structural characteristics of WO_3 and H_xWO_3 , we further examine their potentiometric pH responses. Figure 3a,b shows the pH reversible responses of WO_3 and H_xWO_3 recorded as a function of time within pH range of ≈ 1 –11. Upon increasing the pH, the electromotive force (EMF) signals of both electrodes show a quantitative decrease. As expected, the cyclic pH variations examine the reversible pH sensing. The WO_3 has a poor reversible response with a hysteresis value of up to 71.99 mV (Figure 3a), while the hysteresis value for H_xWO_3 is only 6.70 mV (Figure 3b). Further numerical fittings disclose the slopes for WO_3 ($-25.52 \text{ mV pH}^{-1}$) and ($-16.98 \text{ mV pH}^{-1}$), which significantly deviates from the Nernstian response (Figure 3c). For the H_xWO_3 , the slopes from the cyclic tests reveal near Nernstian responses, -52.52 and $-51.13 \text{ mV pH}^{-1}$ (Figure 3d). We have also fabricated three individual sensors to evaluate the reproducibility (Figure 3e,f). The results show that the H_xWO_3 exhibits excellent reproducibility compared with WO_3 , particularly when pH is over 7. In addition, a single sensor was tested three times to evaluate the repeatability (Figure S3a,b, Supporting Information), in which the H_xWO_3 discloses nearly overlapped potentials at each pH.

As discussed above, the improvement in sensitivity and reproducibility can be attributed to the proton intercalation, which makes it easier for proton transport in the WO_3 lattice.

In continual monitoring, like wearable sensors, fast response time is also an important parameter related to acquiring dynamic changes of physical condition in real time.^[2] Traditionally, the response time is calculated as the time required for sensors to reach 90% of the equilibrium potential value. The response times were measured at representative pH buffer solutions, respectively (Figure 3g,h). The response times for WO_3 in three buffer solutions are 10–45 s (Figure 3g) while the H_xWO_3 shows ≈ 5 s. Such rapid response makes it have great potential in wearable application. Since proton intercalation makes WO_3 exchange H^+ easily, the response time is thus remarkably reduced.

It is necessary to evaluate the selectivity comprehensively. As shown in Figure 4a,b, upon adding $1 \times 10^{-3} \text{ M}$ interfering ions in pH 7.02 buffer solution progressively, the potential of WO_3 changes apparently while no apparent changes are observed for H_xWO_3 . Then, the interfering ions concentration was increased to $10 \times 10^{-3} \text{ M}$ (Figure 4c,d). The potential of WO_3 fluctuates intensely (the maximum drift over 5 mV) while the H_xWO_3 only fluctuated slightly less than 1 mV. It should be noted that the potential of H_xWO_3 changed obviously when adding $10 \times 10^{-3} \text{ M Ca}^{2+}$, due to Britton–Robinson buffer (B-R buffer) containing PO_4^{3-} that can react with Ca^{2+} to form precipitates (not response to Ca^{2+}). In addition, we further evaluated the selectivity using the separation solution method^[48] (Figure 4e,f). When the concentration of interference ions changes from 10^{-1} to 10^{-7} M , the WO_3 responses to all interference ions, while the H_xWO_3 has not apparent response. Therefore, the proton intercalation not only promotes the proton transport in WO_3 , but also proton occupies the lattice, forcing that WO_3 can only exchange with proton to enhance the selectivity.

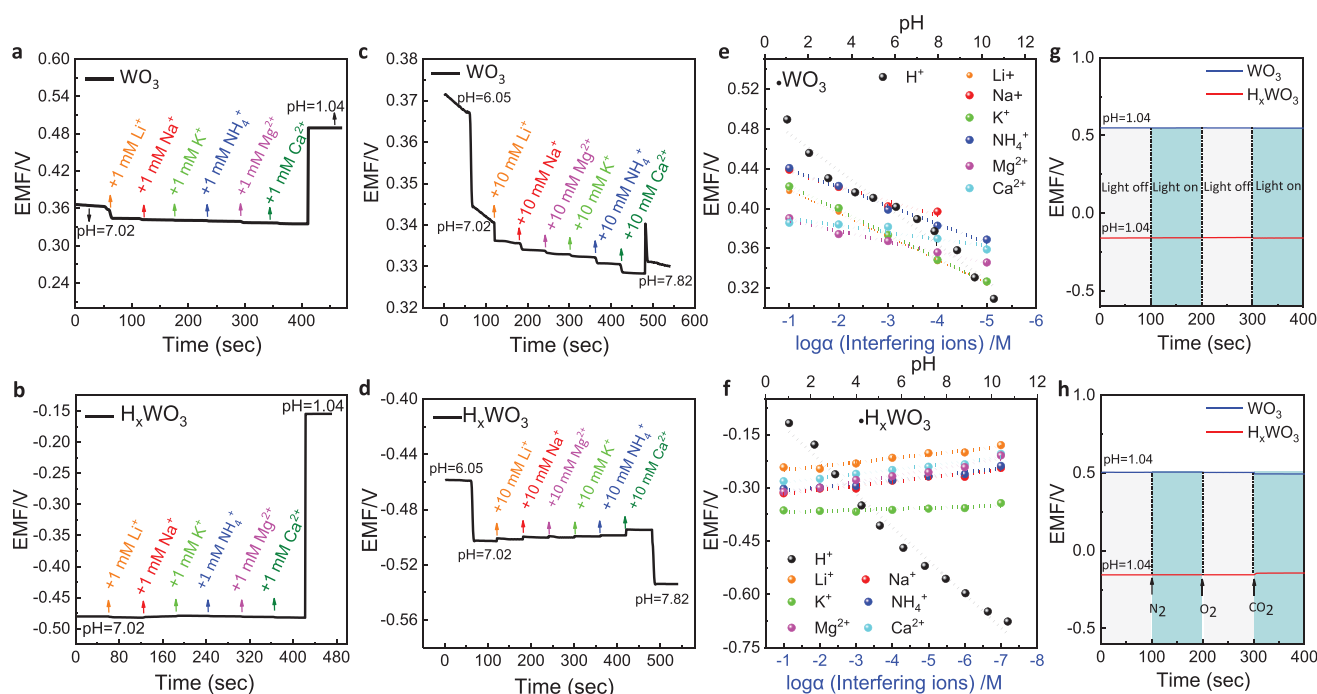


Figure 4. Selectivity and anti-interference for WO_3 and H_xWO_3 . a–d) The selectivity evaluation of WO_3 and H_xWO_3 by continually adding of interfering ions 1×10^{-3} and 10×10^{-3} M. e, f) The selectivity evaluation of WO_3 and H_xWO_3 by separation solution method. g, h) The anti-interferences of WO_3 and H_xWO_3 from light and gas.

It is necessary to examine the resistance to organic interferences in addition to ions since the existence of a variety of organic species in sweat, typically like glucose, ascorbic acid (AA), uric acid (UA), and lactic acid (LA).^[2] As show in Figure S4 in the Supporting Information, the EMF of WO_3 changes apparently toward UA and AA, particularly for highly reductive AA with a potential fluctuation over 100 mV. The reason is that the oxidative WO_3 could oxidize these reductive organic species. H_xWO_3 shows strong resistance to all species (≈ 5 mV fluctuation) except for LA. After proton intercalation, the oxidizing ability was reduced as lowering the valence state of W so that avoiding the interferences of AA and UA. The EMFs for both WO_3 and H_xWO_3 are varied by LA due to the change of solution pH. It was found that the solution showed a pH change from 6.95 to 5.75 before and after addition of LA, thus leading to the change of EMFs. Overall, the proton intercalated H_xWO_3 reveals more anti-interferences ability toward organic species in sweat.

Additionally, the interferences such as light and gas were examined (Figure 4g, h). Light has no effects on both WO_3 and H_xWO_3 (Figure 4g). CO_2 has a slight effect because CO_2 can change the pH value under the saturated concentration (Figure 4h). However, the concentration of CO_2 in the air was not enough to interfere with the potential under normal conditions. Above results indicate good selectivity and anti-interference for the H_xWO_3 .

2.4. Flexible Sensor and Sweat pH Monitoring

A flexible H_xWO_3 -based pH electrode has been designed. Figure 5a, d show the fabrication of a typical chip design containing a reference electrode (RE) and working electrode (WE) (both 5.0 mm in diameter). The detailed fabrication

process is described in experimental section. The microwell pattern of Au electrode substrate was fabricated by magnetron sputtering deposition (Figure 5e). Figure 5f shows the photograph of the flexible sensor after WO_3 modification and RE coating. The solid RE was fabricated by solid Ag/AgCl and solid electrolyte coating on the Au substrate. The proton intercalation was performed by the LSV leading to black H_xWO_3 formed (Figure 5g). Figure 5i demonstrates the good anti-interferences of solid RE toward cations and anions and relatively stability was resulted. We further performed bending tests around a 12 mm diameter teflon cylinder (curving over 270°) (Figure 5h). Figure 5j shows the comparison of potentiometric pH responses between normal and curving states. Nearly overlapped potentials at each pH are resulted. Meanwhile, the response time of normal and curving state both less than 10.5 s and the sensitivity up to -50.54 and -51.77 mV pH^{-1} , respectively (Figure 5k). These results indicated that the prepared flexible pH sensor could be used in the real-time analysis of human sweat.

Finally, the on-body pH sweat analysis was examined (Figure 6a). A sweatband fluidic platform was used, which could be comfortably worn on the forehead of a human subject for efficient sweat collection and wearable analysis during indoor exercise (Figure 6b). Sweat that appears on the subject's forehead can flow effectively into the channel and be transported to the surface of the chip, driven by gravity and motion oscillations (Figure 6a). The prepared flexible pH sensor is bonded with conducting wires, connected to the home-made handed potentiometer with $10^{13} \Omega$ input-resistance and assembled with a battery into a complete wearable sweat sensing platform (Figure 6b). The potentiometer contains a Bluetooth module that monitors the potential changes in real time on the cell phone. To avoid possible potential changes that may

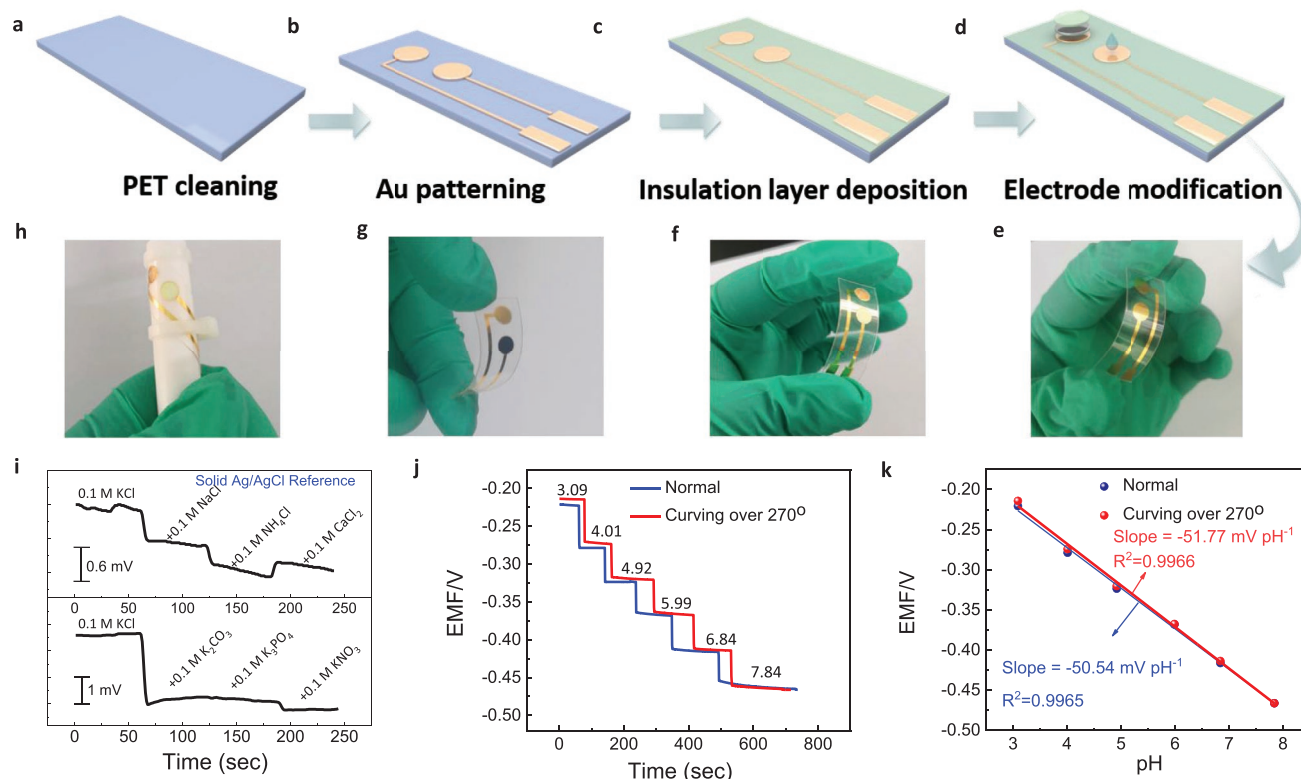


Figure 5. Flexible H_xWO_3 -based pH sensor. a–d) The schematic fabrication of flexible pH sensor including a) PET substrate cleaning by plasmonic, b) Au patterning by sputter, c) insulation layer deposition by PDMS, and d) WE/RE electrodes modification by drop casting. e) The photograph of sputtered flexible Au electrode substrate. f) The photograph of electrodes after surface modifications. g) The photograph of electrode after proton intercalation. h) The photograph of electrode curving for the bending test surrounded on a 12 mm diameter teflon cylinder. i) The ion anti-interference test for the solid Ag/AgCl RE. j) The pH responses for the flexible sensor under normal and curving over 270° states. k) Linear fitting curves under normal and curving states.

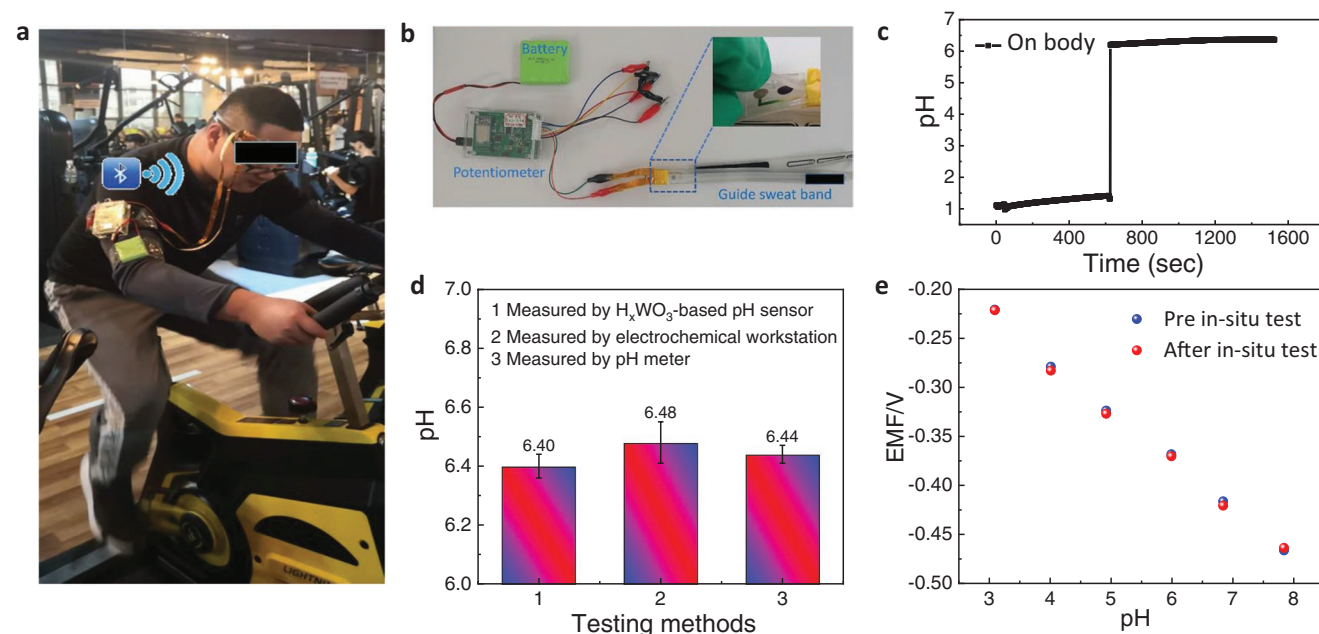


Figure 6. On-body pH monitoring for sweat analysis. a) A photograph illustrates the real-time on-body sweat pH monitoring during indoor cycling. b) Wearable pH sensing device contains battery, potentiometer, pH sensor, and sweat band. c) Real-time sweat pH analysis by the device. d) A comparison of sweat pH measured by ex situ electrochemical workstation and pH meter. e) Calibration curves of the pH sensor before and after sweat test.

affect the signal reading after exposure to sweat sample,^[16] the integrated pH sensor was calibrated with a 0.1 M KCl standard solution and a 0.05 M sulfate solution before and after the on-body trials. As shown in Figure 6a, the subject was cycling at indoor wearing the prepared flexible sweat sensor. After about 11 min, sweat was produced, real-time potentiometric response of pH was shown in Figure 6c. The average of pH from 667 to 1227 s was $\text{pH} \approx 6.40$. To verify the accuracy of the real-time results, a sweat sample was collected following the same exercise process and analyzed by electrochemical workstation and glass pH meter. Similar results were obtained ($\text{pH} = 6.48$ and 6.44) (Figure 6d). Additionally, we also examined the calibration curves of the sensor immediately after the on-body test. There are no significant fluctuations in the potentiometric response, indicating that real-time analysis data is reliable (Figure 6e).

3. Discussion

We have summarized the WO_3 -based solid pH sensors during the past two decades (Table S1, Supporting Information). It has been found significant differences for the analytical performances. The sensitivities were reported from -38.9 to -63.4 mV pH^{-1} . It seems that the response is related with the structures and morphologies of WO_3 . Three representative methods have been proposed for the preparation of WO_3 , including electrochemical oxidation, sputtering, and hydrothermal wet-chemistry. The electrochemical oxidation of W electrode to form a surface thin layer of WO_3 is the origin of proton response. The sensitivity is around -50 mV pH^{-1} (ref. [34,35]). For the sputtering method, the case is complex

in which the sensitivities of -41 to -577 mV pH^{-1} have been reported.^[28,36–38] We also use the sputtering method to prepare WO_3 electrodes with different thicknesses (Figure 7a; Figure S5, Supporting Information). The potentiometric tests demonstrate that the sensitivity decreases with the thickness of WO_3 ($-46.06 \text{ mV pH}^{-1}$ for $0.6 \mu\text{m}$, $-42.98 \text{ mV pH}^{-1}$ for $1.5 \mu\text{m}$, $-39.12 \text{ mV pH}^{-1}$ for $3.9 \mu\text{m}$, and $-24.64 \text{ mV pH}^{-1}$ for $5.3 \mu\text{m}$) (Figure 7b). In addition, the response pH range either decreases with the thickness. For example, the 3.9 and $5.3 \mu\text{m}$ WO_3 exhibits only a linear response between $\text{pH} = 1\text{--}7$ (Figure 7b). This thickness-dependent response might be the reason for the reported different results by sputtering method.

The state of WO_3 on the surface of electrode can be divided into two layers, i.e., surface hydration layer (H_xWO_3) and ion-to-electron transducer layer (bulk WO_3) (Figure 7c), which is similar to the polymer-based solid-contact ion-selective electrodes. The surface hydration layer can be spontaneously formed upon contacting aqueous solution like traditional glass pH electrode, which determines the proton selectivity. The bulk WO_3 layer functions the ion-to-electron transformation. Due to its low electronic conductivity, the thicker WO_3 with high resistance results in the decrease of sensitivity. (1) The sputtered WO_3 , particularly in a crystal state, exhibits a dense film so that the electrolyte cannot diffuse into the bulk layer (Figure 7d). The electrons must transport a dense and high-resistance pathway. The thickness-dependent sensitivity is thus more obvious for dense crystal WO_3 film. The sputtered amorphous WO_3 discloses Nernstian response^[28] because the electrolyte could diffuse into amorphous bulk layer, which results in the formation of inner hydration layer and lowers the resistance. The electrochemical oxidation of W electrode makes amorphous thin-layer WO_3 on the surface, which is also the reason for the observation

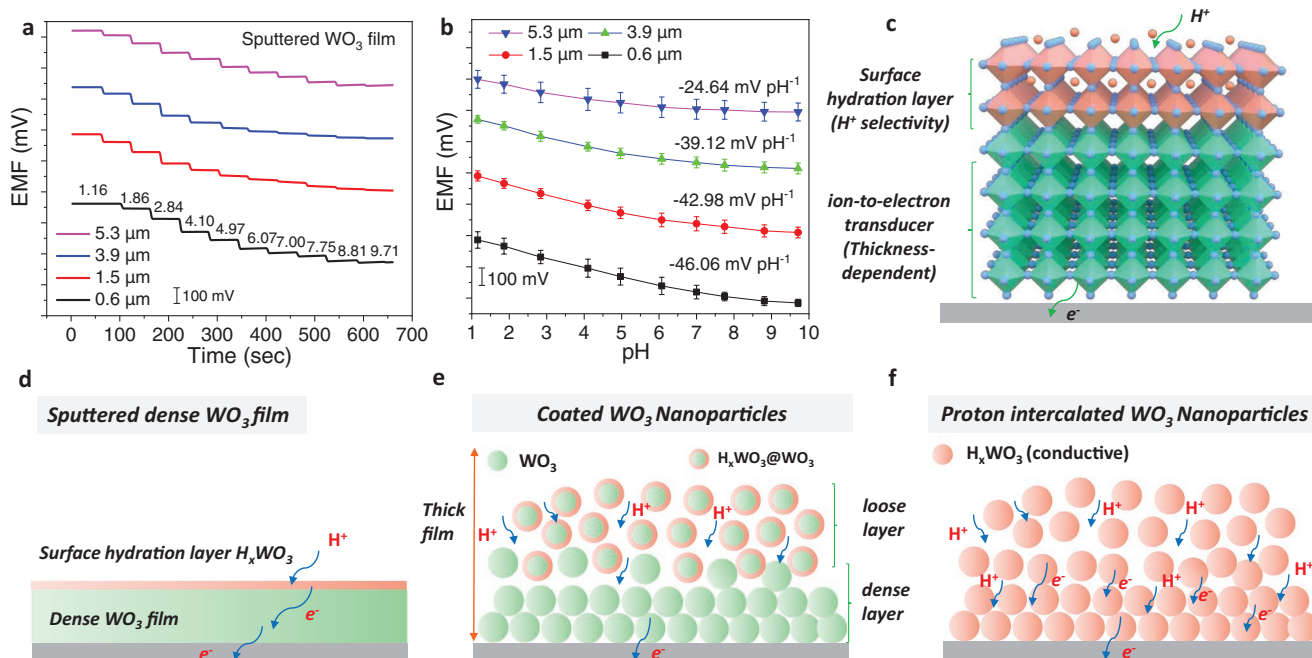


Figure 7. pH response mechanism. a) pH sensors based on sputtered WO_3 films with different thicknesses. b) Corresponding pH calibration curves for the sputtered WO_3 pH sensors. c) pH response mechanism for WO_3 including surface hydration layer and bulk ion-to-electron transducer layer. d–f) pH response models for the d) sputtered WO_3 , e) coated WO_3 nanomaterials, and f) proton intercalated WO_3 nanomaterials.

of Nernstian sensitivity. (2) For the chemical synthesis of WO_3 nanomaterials, it is generally drop-casted or spin-coated on the surface of electrode, which shows at least μm -scale thickness (thick film). The state on the electrode is not such dense like sputtered WO_3 . The structure is constituted by loose layer and inner dense layer (Figure 7e). The WO_3 nanoparticles in loose layer can be surface spontaneously hydrated. However, the inner dense layer with high-resistance leads to low sensitivity (-25.8 mV pH^{-1} in this work and -38.9 mV pH^{-1} in ref. [33]). (3) For the proton intercalated of chemically prepared WO_3 , all WO_3 nanoparticles are transformed into conductive and selective H_xWO_3 by an external electrochemical driving. Therefore, both proton and electron pathways could be easily transferred (Figure 7f). The efficient ion-to-electron transduction enhances the sensitivity and response.

The rapid response is of significance to wearable sensors for the sweat pH monitoring since it can instantly offer the dynamic health information. It should be noted that the proton intercalated WO_3 reveals ultrafast response ($<5 \text{ s}$), which is the fastest response time to the best of our knowledge. The response time is determined by the proton transport and ion-to-electron transduction. For pristine WO_3 , protons can only transport in the surface hydration layer while the H_xWO_3 could efficiently transport protons in the whole material. In addition, the electronic conductive H_xWO_3 further accelerates the ion-to-electron transduction. The hysteresis for the pH loop is only 6.70 mV (pH 1–11) due to fast proton transport, which is either a low value (Table S1, Supporting Information). Significantly, only few MO_x have been used for building flexible pH sensors, and most of them have not been used to detect human sweat (on-body analysis). We have integrated proton intercalated WO_3 on a miniaturized chip and successfully monitored the pH change of sweat during human exercise. Compared to previous MO_x -based flexible pH sensors, the developed wearable sensor exhibits high sensitivity, good flexibility and rapid response time (Table S2, Supporting Information).

4. Conclusions

We have developed an approach of electrochemical lattice H^+ intercalation to regulate the sensitivity and selectivity for WO_3 -based solid-contact pH sensor. This method can remarkably improve the proton exchange capacity of WO_3 sensing membrane and enhance the efficiency of ion-to-electron transduction. The proton intercalated H_xWO_3 exhibits advantages of analytical performances, including reversible response, high selectivity, and fast response time and good anti-interferences. Combining with solid reference electrode, the H_xWO_3 electrode was integrated on a miniaturized chip for wearable sweat pH monitoring. On-body sweat pH analysis disclosed a high accuracy compared with ex situ analysis of the collected sweat. The calibrated results of before and just after on-body measurements indicated that the integrated sensors can offer reliable results for sweat analysis. In future, we will work on using metal oxides to detect other ions in human sweat, and integrating them into miniaturized chips in couple with a temperature-sensitive element to build a complete wearable sweat sensing platform.

5. Experimental Section

Reagents: Sodium tungsten oxide, Lithium sulfate, ammonium sulfate, Magnesium sulfate, sodium sulphate, potassium sulphate, calcium chloride, ferric chloride, nitric acid, sodium chloride, poly(vinylidene fluoride) (PVDF), N-Methyl-2-pyrrolidinone 99.5% (NMP), polydimethylsiloxane (PDMS), bis (2-ethylhexyl) sebacate (DOS), potassium tetrakis (pentafluorophenyl) borate (KTPFB), tetrahydrofuran (THF), methyltridodecylammonium chloride (TDMA-Cl), boracic acid and sodium hydroxide were purchased from Sigma-Aldrich. Nitric acid, sulfuric acid and phosphoric acid were purchased from Sinopharm (China). Silver conductive ink (Jet-600C) was purchased from Hisense. All the chemical reagents and solvents were analytical grade and directly used without further purification. All the aqueous solutions were prepared with ultrapure water. The triacid mixture consists of 0.04 M phosphate, boric acid and acetic acid. Britton–Robinson buffers (B-R buffer) with different pH were prepared by mixing different volumes of the triacid mixture and 0.2 M sodium hydroxide.

Synthesis of WO_3 : WO_3 nanoparticles were prepared by one-step hydrothermal treatment and subsequent post-annealing process. Briefly, 10 mL of 65% nitric acid and 0.989 g $\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O}$ were dissolved in 50 and 10 mL of ultrapure water under constant magnetic stirring, respectively. Then a yellow precursor solution was formed by mixing the above two solutions. Subsequently, the mixed solution was transferred into a 100 mL Teflon lined stainless autoclave and heated at 180°C for 3 h . After reaction, the obtained yellow precipitate was washed using ultrapure water and ethanol 5 times, and then dried at 60°C . The WO_3 nanoparticles were finally obtained by annealing the products at 350°C for 3 h . The WO_3 thin films were fabricated by magnetron sputtering deposition with 50 mm diameter W target (AJA Orion 5). The fluorine-doped tin oxide was used as substrates. The magnetron was operated with a power of 120 W for about 0.57 , 1.43 , 2.86 , and 9.4 h respectively in Ar/O_2 atmosphere with the volume flow ratio of $5:9$. The pressure during deposition was fixed at 3 mtorr and sputtering temperature was 500°C . The thickness of WO_3 film was obtained by the SEM (Phenom LE, Netherlands).

Material Characterizations: The crystallographic structure of WO_3 film was observed by X-ray diffraction (XRD, PANalytical X'Pert Powder, Netherlands) using $\text{Cu K}\alpha$ radiation ($\lambda = 1.54056 \text{ \AA}$), and the detector scanned over a 2θ range of angles from 20° to 80° with a step size of 0.02° . The morphologies and structures were examined using the SEM. Raman scattering spectra at room temperature were performed by a 200 mW argon-ion laser microscope (LabRAM HR Evolution, France) at an excitation wavelength of 532 nm . X-ray photoelectron spectroscopy (XPS) data were obtained by X-ray photoelectron spectrometer (ThermoScientific ESCALAB 250Xi, USA) $\text{Al K}\alpha$ X-ray radiation as the X-ray source for excitation.

Fabrication of WO_3 and H_xWO_3 -Based pH Sensor: A stable dispersion liquid of sensing materials was obtained by dispersing 50 mg WO_3 nanoparticles and 12.5 mg PVDF in $675 \mu\text{L}$ NMP and sonicated for 3 h . The WO_3 -based pH sensor was prepared by depositing $10 \mu\text{L}$ of the above dispersion liquid on a well-polished glass carbon (GC) substrate (diameter 5 mm), and the membranes were then dried in air at 50°C . Subsequently, the H_xWO_3 -based electrode was obtained by using linear sweep voltammetry (LSV from 0 to -0.45 V with scanning rate of 0.5 mV s^{-1}) to insert H^+ into WO_3 electrode. The prepared WO_3 and H_xWO_3 pH sensors were conditioned in 0.05 M H_2SO_4 solution for 12 and 2 h before measurements, respectively.

Electrochemical Measurements: All electrochemical measurements were performed by using a Gamry workstation (Reference 600+, USA) and multi-channel potentiometer EMF6 (Lawson Lab, Inc) at room temperature. The WO_3 or H_xWO_3 is the working electrode (WE). Reference electrode and counter electrode are saturated calomel electrode (SCE) and Pt electrode, respectively. The electromotive force (EMF) between changes between the working and reference electrodes were recorded in different pH levels in the range of pH 1–11. Solutions

with different pH levels from B-R buffer solution prepared by mixing 0.04 M three acid mixed liquids and 0.2 M NaOH solution with pH range of 2–11. Electrochemical impedance spectra (EIS) study was recorded in the frequency range 1 MHz–0.1 Hz with an open circuit voltage and an excitation amplitude of 10 mV.

Fabrication of Flexible pH Sensor: First, the polyethylene terephthalate (PET) membrane was cleaned by using acetone, isopropanol and O₂ plasma etching. The microwell pattern of Au electrode was fabricated by magnetron sputtering deposition with Au target at a power of 60 W for 25 min in Ar atmosphere. Then, PDMS insulating layer was deposited by using Spinner. Lastly, WO₃ working electrode was fabricated on the obtained Au electrode as the case on the GC electrode. Before use, the flexible WO₃ electrode was transformed into H_xWO₃ by the same LSV for proton intercalation like on the GC electrode. The solid reference electrode was fabricated as follows.

Fabrication of PVC/KCl-Ag/AgCl Reference Electrode: First, the Ag electrode was prepared by depositing 10 µL of silver conductive ink on Au substrate (diameter 5 mm), and the membranes were then dried in vacuum at 140 °C for 20 min. Then, the Ag/AgCl reference electrodes were obtained by injecting 10 µL 0.1 M FeCl₃ solution on top of each Ag reference electrode using a micropipette for 1 min. The reference electrode solution was prepared by dissolving KTPFB (0.9% weight by weight, w/w), TDMA-Cl (1.1% w/w), DOS (68% w/w) and PVC (30% w/w) in THF. Then, 40.4 mg KCl and 15.3 mg AgCl powers were added to 250 µL equal parts of the above solution. The reference protective layer cocktail formed by dissolving 4 g of PVC (33.1% w/w) and DOS (66.9% w/w) in 50 mL THF. Finally, the PVC/KCl-Ag/AgCl reference electrode was obtained by depositing 10 µL above two solution respectively. The dropping time interval of the two solutions was over 12 h.

On-Body Sweat pH Analysis: The prepared flexible electrode was assembled into a complete wearable sweat sensing sensor. A healthy male volunteer (age 27, weight 80 kg) wore the sensor and rode indoors at a steady pace until sweat was produced. The home-made mini-potentiometer with input resistance of 10¹³ Ω was used to record the EMF for on-body pH monitoring. After in-situ tests, the sweat samples were collected, and ex situ analyzed by Gamry workstation and pH meter, respectively. The informed signed consent was obtained from the volunteer engaging in the activity of this study.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

proton intercalation, solid-contact pH sensor, sweat analysis, wearable sensor, WO₃

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- [1] M. T. Ghoneim, A. Nguyen, N. Dereje, J. Huang, G. C. Moore, P. J. Murzynowski, C. Dagdeviren, *Chem. Rev.* **2019**, *119*, 5248.
- [2] M. Bariya, H. Y. Y. Nyein, A. Javey, *Nat. Electron.* **2018**, *1*, 160.
- [3] J. Kim, A. S. Campbell, B. E.-F. de Ávila, J. Wang, *Nat. Biotechnol.* **2019**, *37*, 389.
- [4] J. R. Sempionatto, I. Jeerapan, S. Krishnan, J. Wang, *Anal. Chem.* **2020**, *92*, 378.
- [5] N. Uriá, N. Abramova, A. Bratov, F.-X. Muñoz-Pascual, E. Baldrich, *Talanta* **2016**, *147*, 364.
- [6] E. M. Briggs, S. Sandoval, A. Erten, Y. Takeshita, A. C. Kummel, T. R. Martz, *ACS Sens.* **2017**, *2*, 1302.
- [7] H. J. Chung, M. S. Sulkin, J. S. Kim, C. Goudeseune, H. Y. Chao, J. W. Song, S. Y. Yang, Y. Y. Hsu, R. Ghaffari, I. R. Efimov, J. A. Rogers, *Adv. Healthcare Mater.* **2014**, *3*, 59.
- [8] A. N. Bezbaruah, T. C. Zhang, *Anal. Chem.* **2002**, *74*, 5726.
- [9] Y. J. Lin, M. Bariya, A. Javey, *Adv. Funct. Mater.* **2020**, 2008087.
- [10] Y. H. Qin, A. U. Alam, M. M. R. Howlader, N. X. Hu, M. J. Deen, *Adv. Funct. Mater.* **2016**, *27*, 4923.
- [11] W. Gao, H. Ota, D. Kiriya, K. Takei, A. Javey, *Acc. Chem. Res.* **2019**, *52*, 523.
- [12] J. B. Tu, R. M. Torrente-Rodriguez, M. Q. Wang, W. Gao, *Adv. Funct. Mater.* **2020**, *29*, 1906713.
- [13] H. Y. Y. Nyein, W. Gao, Z. Shahpar, S. Emaminejad, S. Challa, K. Chen, H. M. Fahad, L.-C. Tai, H. Ota, R. W. Davis, A. Javey, *ACS Nano* **2016**, *10*, 7216.
- [14] Y. Lin, M. Bariya, H. Y. Y. Nyein, L. Kivimäki, S. Uusitalo, E. Jansson, W. Ji, Z. Yuan, T. Happonen, C. Liedert, J. Hiltunen, Z. Fan, A. Javey, *Adv. Funct. Mater.* **2019**, *29*, 1902521.
- [15] W. Gao, S. Emaminejad, H. Y. Y. Nyein, S. Challa, K. Chen, A. Peck, H. M. Fahad, H. Ota, H. Shiraki, D. Kiriya, D.-H. Lien, G. A. Brooks, R. W. Davis, A. Javey, *Nature* **2016**, *529*, 509.
- [16] M. Parrilla, M. Cuartero, G. A. Crespo, *TrAC, Trends Anal. Chem.* **2019**, *110*, 303.
- [17] W. Y. He, C. Y. Wang, H. M. Wang, M. Q. Jian, W. D. Lu, X. P. Liang, X. Zhang, F. C. Yang, Y. Y. Zhang, *Sci. Adv.* **2019**, *5*, eaax0649.
- [18] Y. Lyu, S. Gan, Y. Bao, L. Zhong, J. Xu, W. Wang, Z. Liu, Y. Ma, G. Yang, L. Niu, *Membranes* **2020**, *10*, 128.
- [19] Y. Shao, Y. Ying, J. Ping, *Chem. Soc. Rev.* **2020**, *49*, 4405.
- [20] E. Zdrachek, E. Bakker, *Anal. Chem.* **2021**, *93*, 72.
- [21] J. Xu, Z. Zhang, S. Gan, H. Gao, H. Kong, Z. Song, X. Ge, Y. Bao, L. Niu, *ACS Sens.* **2020**, *5*, 2834.
- [22] L. E. Ibarra, L. Tarres, S. Bongiovanni, C. A. Barbero, M. J. Kogan, V. A. Rivalola, M. L. Bertuzzi, E. I. Yslas, *Ecotox. Environ. Safe.* **2015**, *114*, 84.
- [23] M. Odowd, M. Martin, A. Wheble, M. D. G. Gillmer, P. Rolfe, *J. Biomed. Eng.* **1988**, *10*, 165.
- [24] P. Chao, D. Ammann, U. Oesch, W. Simon, F. Lang, *Pfluegers Arch.* **1988**, *411*, 216.
- [25] J. Y. Jeon, B. C. Kang, T. J. Ha, *Appl. Surf. Sci.* **2020**, *514*, 145956.
- [26] M. Ternois, M. Mougou, E. Flahaut, A. Dussutour, *Nanotoxicology* **2021**, *15*, 511.

- [27] L. Manjakkal, D. Szwagierczak, R. Dahiya, *Prog. Mater. Sci.* **2020**, 109, 100635.
- [28] P. Salazar, F. J. Garcia-Garcia, F. Yubero, J. Gil-Rostra, A. R. González-Elipé, *Electrochim. Acta* **2016**, 193, 24.
- [29] L. Santos, J. P. Neto, A. Crespo, D. Nunes, N. Costa, I. M. Fonseca, P. Barquinha, L. Pereira, J. Silva, R. Martins, E. Fortunato, *ACS Appl. Mater. Interfaces* **2014**, 6, 12226.
- [30] Y. R. Cheng, F. Yang, G. L. Xiang, K. Zhang, Y. Cao, D. D. Wang, H. F. Dong, X. J. Zhang, *Nano Lett.* **2019**, 19, 1179.
- [31] S. J. Yuan, H. He, G. P. Sheng, J. J. Chen, Z. H. Tong, Y. Y. Cheng, W. W. Li, Z. Q. Lin, F. Zhang, H. Q. Yu, *Sci. Rep.* **2013**, 3, 1315.
- [32] C.-Y. Kuo, S.-J. Wang, R.-M. Ko, H.-H. Tseng, *JPN J. Appl. Phys.* **2018**, 57, 04FM09.
- [33] S.-J. Choi, S. Savagatrup, Y. Kim, J. H. Lang, T. M. Swager, *ACS Sens.* **2019**, 4, 2593.
- [34] K. Yamamoto, G. Shi, T. Zhou, F. Xu, M. Zhu, M. Liu, T. Kato, J.-Y. Jin, L. Jin, *Anal. Chim. Acta.* **2003**, 480, 109.
- [35] S. Drenslar, S. Walkner, C. C. Mardare, A. W. Hassel, *Phys. Status. Solidi. A* **2014**, 211, 1340.
- [36] W.-D. Zhang, B. Xu, *Electrochem. Commun.* **2009**, 11, 1038.
- [37] J.-L. Chiang, S.-S. Jan, J.-C. Chou, Y.-C. Chen, *Sens. Actuators, B* **2001**, 76, 624.
- [38] A. Lale, A. Tsopela, A. Cívelas, L. Salvagnac, J. Launay, P. Temple-Boyer, *Sens. Actuators, B* **2015**, 206, 152.
- [39] H. A. Evans, Y. Wu, R. Seshadri, A. K. Cheetham, *Nat. Rev. Mater.* **2020**, 5, 196.
- [40] S. Tanisaki, *J. Phys. Soc. Jpn.* **1960**, 15, 573.
- [41] Y. Cui, F. Liang, C. Ji, S. Xu, H. Wang, Z. Lin, J. Liu, *ACS Omega* **2019**, 4, 7428.
- [42] J.-H. Ha, P. Muralidharan, D. K. Kim, *J. Alloys Compd.* **2009**, 475, 446.
- [43] L.-L. Xing, K.-J. Huang, L.-X. Fang, *Dalton. Trans.* **2016**, 45, 17439.
- [44] A. W. Sleight, J. L. Gillson, *Solid State Commun.* **1966**, 4, 601.
- [45] P. G. Dickens, R. J. Hurditch, *Nature* **1967**, 215, 1266.
- [46] S. Han, J. Li, X. Chen, Y. Huang, C. Liu, Y. Yang, W. Li, *Int. J. Hydrogen Energy* **2012**, 37, 16810.
- [47] Y. Ren, Y. Zou, Y. Liu, X. Zhou, J. Ma, D. Zhao, G. Wei, Y. Ai, S. Xi, Y. Deng, *Nat. Mater.* **2020**, 19, 203.
- [48] E. Bakker, E. Pretsch, P. Bühlmann, *Anal. Chem.* **2000**, 72, 1127.